

Shell Scripting

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Lab-1

Using sed utility:

- 1- Display the lines that contain the word “lp” in /etc/passwd file.
`sed -n '/lp/p' /etc/passwd`
- 2- Display /etc/passwd file except the third line.
`sed '3d' /etc/passwd` or `sed -n '3!p' /etc/passwd`
- 3- Display /etc/passwd file except the last line.
`sed '$d' /etc/passwd` or `sed -n '$!p' /etc/passwd`
- 4- Display /etc/passwd file except the lines that contain the word “lp”.
`sed '/lp/d' /etc/passwd` or `sed -n '/lp!p' /etc/passwd`
- 5- Substitute all the words that contain “lp” with “mylp” in /etc/passwd file.
`sed 's/lp/mylp/g' /etc/passwd`

Using awk utility:

- 1- Print full name (comment) of all users in the system.
`awk -F: '{ print $5 }' /etc/passwd`
- 2- Print login, full name (comment) and home directory of all users.(Print each line preceded by a line number)
`awk -F: '{ print NR, $1, $5, $6 }' /etc/passwd`
- 3- Print login, uid and full name (comment) of those uid is greater than 500
`awk -F: '$3 > 500 { print $1, $3, $5 }' /etc/passwd`
- 4- Print login, uid and full name (comment) of those uid is exactly 500
`awk -F: '$3 == 500 { print $1, $3, $5 }' /etc/passwd`
- 5- Print line from 5 to 15 from /etc/passwd
`awk -F: 'NR >= 5 && NR <= 15 { print $0 }' /etc/passwd`
- 6- Change lp to mylp
`awk '{ gsub(/lp/, "mylp"); print $0 }' /etc/passwd`

7- Print all information about greatest uid.

```
awk -F: 'BEGIN { max=0 } $3 > max { max=$3; line=$0 } END { print line }'
/etc/passwd
```

8- Get the sum of all accounts id's

```
awk -F: '{ sum += $3 } END { print sum }' /etc/passwd
```

Bonus:

1. Get the sum of accounts id's that has the same group.

```
awk -F: 'BEGIN { print "GID\tUID_SUM" } { sum[$4] += $3 } END { for (gid in sum) print gid "\t" sum[gid] }' /etc/passwd
```

2. Make the following report: User-Group Report

```
awk -F: '
BEGIN {
    print "User-Group Report"
    print "-----"
}
{ groups[$4] = groups[$4] $1 "\n" }
END {
    for (gid in groups) {
        print "Group ID " gid ":"
        print groups[gid]
    }
}' /etc/passwd
```